

产品手册-PRODUCT BROCHURE

VCC: 真空浇注干式变压器

VCC: Cast coil dry-type transformers



ABB全球

ABB worldwide

几乎在人们生活工作的每个地方，都至少会找到一个变压器。但只要它一直工作着，向百货商场的自动扶梯、酒店电梯、办公室计算机、本地面包店的烤炉或石油化工厂提供电力，没有人会对其再多考虑。ABB是电力与自动化技术领域的全球领先企业，与全世界的公共设施和工业客户合作，致力于使其降低环境影响的同时提高业绩。作为世界领先的工程公司之一，ABB帮助客户以可持续方式有效利用电力和提高生产效率。

In almost every place where people live and work you will find at least one transformer. But as long as it keeps working and supplying power to the escalator in the department store, the hotel lift, the office computer, the oven in the local bakery or the petrochemical plant, no one gives it a second thought. ABB is a global leader in power and automation technologies enabling utility and industry customers from around the world to improve performance while lowering environmental impact. As one of the world's leading engineering companies, ABB helps its customers to use electrical power effectively and to increase industrial productivity in a sustainable way.



为何选择干式：安全且环保的变压器

- 降低环境污染
- 易燃或有毒物质的零泄漏风险
- 环境友好型生产过程
- 极适用于排放和污染区域
- 阻燃、自熄
- 抗短路能力强
- 过载能力强
- 高度地震设防
- 能承受最恶劣的振动工况

Why choose dry-type: safe and environmental friendly transformers

- Reduced environmental contamination
- Zero risk of leakage of flammable or contaminating substances
- Environmentally friendly production
- Well suited to dump and contaminated areas
- Non flammable and self-extinguishing
- High resistance to short circuits
- High capacity to support overloads
- High performance in dealing with seismic phenomenon
- Capable of withstanding the most severe rolling and vibrating conditions

最新进展: 扩展产品组合

HiDry⁷², 能达到 63 MVA且工作电压可高达72.5 kV。本产品可大大节约以下方面的成本: 土建工程、消防系统、保险费用、现场安装、电缆铺设、维护。

ABB hi-T Plus 变压器是一款具有更高绝缘耐热等级 (H 级: 180°C) 的优良产品。好处是提高了绝缘寿命和过载能力, 从而可为您实现更为优化的设计。

EcoDry: 此极高效的新型干式变压器是基于低空载损耗与优异效率的概念而设计的, 适用于需要评估环境影响和总运营成本的客户。

所有此类变压器均可经过特殊设计、定制, 并可根据需要提供各种附件。

Latest developments: expanding the portfolio

HiDry⁷², able to reach 63 MVA and operating voltages up to 72.5 kV. This product offers savings on: civil works, fire systems, insurance fees, site installation, shorter cables, maintenance.

ABB hi-T Plus transformer is a superior product with upgraded thermal insulation level (class H: 180°C). Giving you the benefits of an increased insulation lifetime and overloading capability which allows for a more optimised design.

EcoDry: the new ultra efficient dry-type transformers designed with reduced losses and superior efficiency, for customers evaluating environmental impact and total operational costs.

All these transformers can be designed, customised and supplied with a wide variety of accessories as required.



适用于所有应用领域的可靠解决方案

Reliable solutions for all applications

许许多多的应用领域需要高安全性能、节约成本和改善环境方面的技术。

ABB专业的变压器生产技术能够生产适用于最佳空间利用、特殊要求和最苛刻条件的产品。

ABB电力技术领域的全球领先企业, 在许多应用环境、行业范围和定制化项目方面积累了丰富的经验:

- 风力发电
- 太阳能
- 船舶
- 铁路
- 变频传动
- 发电厂
- 公共设施
- 建筑物
- 化工和制药
- 金属和采矿业
- 纸浆和造纸业
- 石油天然气
- 水泥业
- 城市供水
- 工业
- 基础设施
- 核能

A large variety of applications demand technologies which contribute to high safety performance, cost savings and environmental respect.

ABB has expertise in producing transformers for optimum space utilization, special requirements and the most demanding conditions.

ABB is the global leader in power technologies, providing the broadest experience in all applications, ranges and customized projects:

- Wind
- Solar
- Marine
- Railways
- Drives
- Power plants
- Utilities
- Building
- Chemical and pharmaceutical
- Metal and mining
- Pulp and paper
- Oil and gas
- Cement
- Water
- Industry
- Infrastructure
- Nuclear



我们的技术: 什么使我们与众不同?

Our technology: what makes us different?

ABB采用最先进的生产技术和最严格的控制体系, 以保证最高的产品质量和产品总体可靠性。

- 铁芯采用全斜接缝步进叠片的技术, 可保证最佳性能和最低噪声级。硅钢片自动剪切自动叠放。这样可以保证整叠硅钢片的尺寸精确度和片间搭接。
- 高压绕组采用的是由铜带和双层绝缘连续绕制的饼式结构。绕组浇注是在真空状态下用环氧树脂浇注。绕组电场强度经过瞬态梯度分布分析, 符合系统设计的要求。
- 低压绕组由铜箔和预浸环氧树脂的绝缘纸一起绕制而成。线圈绕制完成后, 会将线圈放入烘箱中固化, 从而形成能承受短路时电动力的坚固致密的线圈。

浇注过程是制造过程中的一个关键阶段。浇注是在最严格的条件下进行, 以确保最佳的绝缘和机械特性。树脂混合物是在受到连续监控的混合罐中进行混合的, 组分仅在浇注过程之前才混合在一起。在真空浇注室内, 树脂注入模具中。这样可确保注入模具时树脂的粘性最小提高缝隙填充效果以保证完成线圈的局部放电电量最小, 从而填充缝并使完成的绕组可达到局部放电的最低等级。

在ABB变压器工厂中, 整个报价、设计和生产过程由精细的软件控制和计划。这样可在保证最高质量标准的同时确保最高的生产效率和最短的生产周期。



ABB uses the most advanced production technologies and the most demanding control systems to guarantee the highest product quality and total product reliability.

- The magnetic core has a miter step joint to ensure optimum performance and minimum sound levels by using step lap technology. The magnetic steel is cut to length sequentially and automatically stacked. This ensures dimensional accuracy and single sheet interlacing within the full stack.
- The high voltage winding has a continuous drop down disc with a strip conductor (copper) and double layer insulation. Windings are cast under vacuum with epoxy resin. Transient analysis tests have been performed to verify the electrical stress distribution through the windings confirming the highest strength in our design.
- The low voltage windings are made of conductor foil (copper), and insulating foil pre-impregnated with epoxy resin. After the winding process the coil is cured into an oven resulting in an extremely compact winding which can withstand the dynamic stresses produced by a short-circuit.

The encapsulation process is one of key stages in manufacture. Encapsulation is carried out under the most rigorous conditions in order to ensure optimum insulating and mechanical characteristics. The resin mix is prepared in a continuously monitored mixing plant where all the components are mixed together just before the encapsulation process. Inside the vacuum casting chamber, the resin is introduced into the mould. The components are mixed together just before the encapsulation process, this ensures that the viscosity of the resin when poured in the moulds is very low, filling interstices and allowing the finished winding to reach the lowest level of partial discharge. Once the casting is finished the coils are placed into the curing oven for the resin gel to cure, and acquire the optimal final electrical and mechanical properties.

In ABB transformer factories, the whole tendering, design and production processes are controlled and planned with sophisticated software. This ensures the highest productivity and reduced production time while maintaining the highest quality standards.

我们的价值

Our values

所有工艺均通过最严格质量标准的认证。

设计

- 保证绕组中的正确温升; 即使在配置IP/NEMA标准外壳的情况下, 也不能降低绝缘耐热等级。
- 可在最低-25°C的环境温度下工作
- 可配置各种附件
- 绝缘等级F、H (选项)

技术

- 高压绕组中选用铜带
- 低压绕组中选用整张铜箔
- 全真空浇注
- 局部放电量<10 pC
- 表面光滑
- 自熄性
- 全密封线圈的防尘性
- 不含硅
- 低磁通密度及步进叠片结构可降低噪声
- 承受雷电冲击能力强
- 抗短路能力强 (径向和轴向)

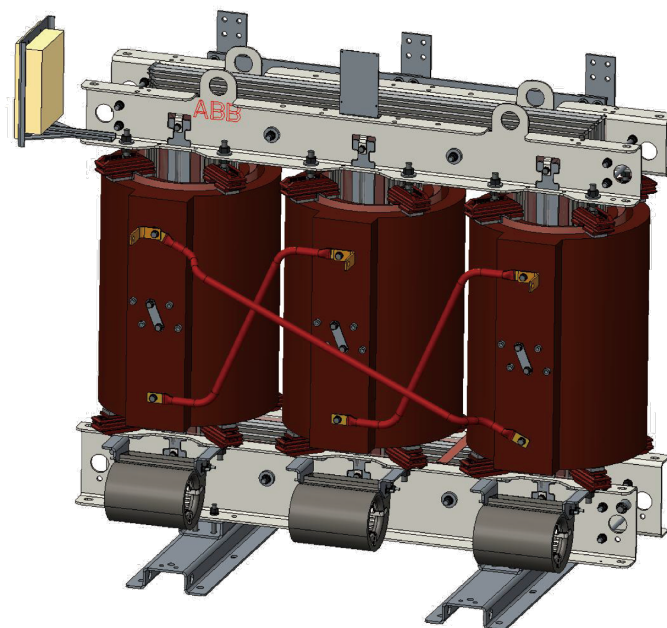
All this process is certified under the strictest quality standards.

Design

- Guarantee of accurate temperature rise in windings; in case of enclosure IP/NEMA there is no de-rating
- Low working temperatures down to -25°C
- Possibility to include any kind of accessories
- Insulation class F, H (option)

Technology

- Copper foil disk in high voltage winding
- Copper foil full width in low voltage winding
- Casting under vacuum
- Low partial discharge values <10 pC
- Smooth surface
- Self extinguishing
- Dust resistance thanks to sealed coils
- Silicon free
- Low induction level and step lap configuration granting lower noise level
- High impulse voltage withstand
- High short circuit withstand (radial and axial)



定制化解决方案

Customized solutions

设计选项

- 低损耗变压器
- 铜绕组
- 低压变压器
- H级变压器
- 低温升
- 多组合初级电压
- 多组合二次绕组
- 多绕组变压器（同一铁芯上有2、3、4或5个）
- 浇注式低压绕组
- 端封式或浇注式低压绕组
- 浸渍式高压绕组
- 灵活的高压和低压出线位置：顶部和底部，侧方出线
- 特殊连接组别
- 自藕变压器
- 适用于6、12、18和24脉冲的变频驱动装置（VSD）、整流器和励磁变压器
- 0.15至0.9的不同耦合系数
- 接地变压器
- 地震设防与防振措施
- 户外运行
- 提供防燃弧罩壳

Design options

- Reduced loss transformers
- Copper windings
- Low voltage transformers
- Class H transformers
- Reduced temperature rise
- Multiple primary voltage
- Multiple secondary windings
- Multi winding transformers (2, 3, 4 or 5 on the same core)
- Encapsulated low voltage windings
- Sealed or casted low voltage winding
- Impregnated high voltage windings
- Different location of high and low voltage connection terminals: top and bottom, both on the same side
- Special connection groups
- Autotransformer
- Variable speed drives (VSD), rectifiers and excitation transformers for 6, 12, 18 and 24 pulses
- Different coupling factors from 0.15 up to 0.9
- Earthing transformers
- Seismic and vibration reinforcement system
- Outdoor operation
- Enclosure passed the test of the arc



定制化解决方案

Customized solutions

附件

- 温度控制器
- 减振器
- 加热器
- 静电屏蔽
- 初级绕组与二级绕组上的电流互感器
- 插拔式套管头
- 高压接地开关
- 避雷器（对于高压和低压）
- 最高可增大50 %功率的冷却风扇
- 有载调压开关（OLTC）
- 软连接
- 接线盒
- 双向轮
- 不同防护等级的外壳以配合电缆或密集母线联接
- 接地球
- 提升与牵拉吊环
- 带电指示器
- 电压自动调整器（VAR）
- 特殊包装
- 水冷却器
- 电缆接线箱
- 滑道

Accessories

- Temperature monitor
- Antivibration pads
- Space heaters
- Electrostatic screen
- Current transformers on primary and secondary windings
- Plug in bushings
- High voltage earthing switch
- Surge arrestors (for high and low voltage)
- Cooling fans with up to 50 % power increase
- On load tap changers (OLTC)
- Flexible terminals
- Connection box
- Bidirectional wheels
- Different enclosures with cable or bus-duct connection
- Earthing bullets
- Lifting and pulling lugs
- Voltage detectors
- Voltage automatic regulation (VAR)
- Special packings
- Hydrocoolers
- Cable boxes
- Skids

测试与物流

Testing and logistics

所有变压器均按照最严格的质量控制标准制造, 以保证产品的整体可靠性。

All transformers are manufactured to the most stringent quality control standards in order to guarantee the full reliability of the product.

常规测试

所有变压器均经过100%的常规测试:

- 电压比测量和相位检查
- 工频耐压测试
- 感应耐压测试
- 局部放电测量
- 空载损耗和电流测量
- 绕组电阻测量
- 负载损耗与短路阻抗测量
- 绝缘电阻测量

型式试验

根据客户要求:

- 温升测试
- 雷电冲击测试

特殊测试

也根据客户要求:

- 噪声级别测试
- 测量零序阻抗
- 测量空载电流的谐波
- 防腐保护检查
- 短路测试

此外, 我们的变压器还具有以下认证:

- F1级“防火性能”认证
- C1和C2级“耐气候性能”认证
- E2级“冷凝和湿度性能”认证
- “具有在最严苛环境条件中运行的资格”

Routine test

All transformers are 100% routine tested:

- Voltage ratio measurement and check of phase displacement
- Separate source voltage-withstand test
- Induced over voltage withstand test
- Partial discharge measurements
- No load loss and current measurement
- Measurements of windings resistances
- Load loss and short circuit impedance measurements
- Measuring insulation resistance

Type tests

Performed upon customer requirements:

- Temperature rise test
- Lightning impulse test

Special tests

Also performed according to customer request:

- Noise level test
- Measuring zero-sequence impedance
- Measuring of harmonics of the no-load current
- Anti-corrosion protection checking
- Short circuit test

Additionally, our transformers have the following certificates:

- The class F1 "Fire behavior" certificate
- The class C1 and C2 "Climatic" certificate
- The class E2 "Condensation and humidity" certificate
- Qualified to perform in the most severe environmental conditions



按照相关的IEC、GB、EN和/或IEEE / ANSI标准和Gost (*)。

According to the relative IEC, EN and/or IEEE/ANSI standards and Gost (*).

测试与物流

Testing and logistics

ABB拥有符合UNE-EN-ISO/IEC 17025:2000标准的认证实验室。

此资格认证赋予认证公司权利, 使其可作为独立的官方实验室进行测试, 并以独立第三方的身份发布相应的测试报告。

物流和服务

变压器成功通过测试后, 产品即可经由陆路或海上发运。

作为制造商, ABB明白物流代表我们直接面向客户; 这就是ABB发展具备相同价值观与发展目标的合格的大型供应商网络的原因之一。

ABB会妥善保管所有的正式文件, 这根据最终目的地和交付方式要求而定。

用于特殊应用或条件的不同包装

- 标准包装
- 板条箱包装
- 适航包装

由于工厂遍布全球, ABB的生产地距离您的安装现场可以很近。若有要求, ABB在安排特殊运输方面也具有多年的经验。

而且, ABB遍布全球的各个部门可确保最佳的本地售后服务。

与我们携手合作, 您可以利用工厂与设施的全球网络, 而且我们会通过一套完整的产品与解决方案为您提供周到的本地服务。我们的质保提供“一个简单的ABB”品质和服务。与我们合作, 可使您享受到各种生产设施的优势, 它们采用最新技术, 符合标准, 并为特殊产品与解决方案提供最高品质。

(*) EC: 国际电工委员会。

EN: 欧洲标准, 概括了所有相关的欧洲标准NF、BS、VDE、DIN、CEI、UNE

ABB has one certified test laboratory according to the standard UNE-EN-ISO/IEC 17025:2000.

This accreditation gives the authority to the certified company to act as an independent official laboratory to test and issue the corresponding test reports as an independent third party.

Logistic and service

Once the transformer is successfully tested; the product is ready for shipping either by truck or sea-freight.

As a manufacturer, ABB understands the relevance that logistics represents to our customers; this is one of the reasons why ABB has developed a large and qualified network of suppliers that have the same values and targets.

ABB takes care of all official documentation, depending on final destinations and delivery terms.

Different packaging for special applications or conditions

- Standard packing
- Crate packing
- Seaworthy packing

With a global factory footprint ABB can produce close to your installation site. ABB also has a long experience in arranging special transport when it is required.

In addition, ABB's global presence ensures the best, local, after-sales service.

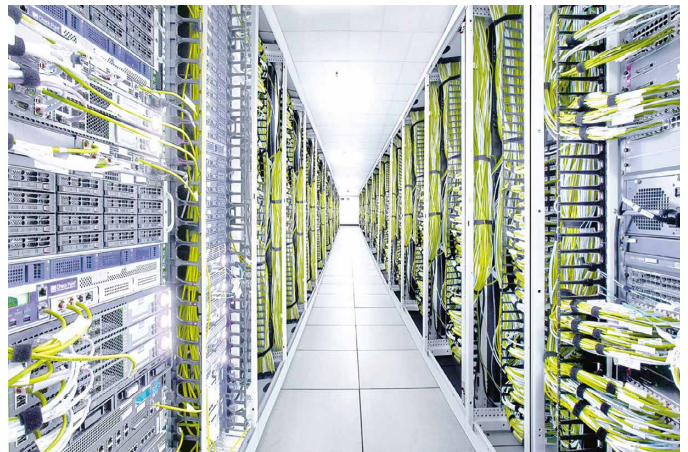
Working with us, you have access to a worldwide network of factories and facilities, serving you locally with a full range of products and solutions. Our warranty provides "One Simple ABB" quality and service. Working together gives you access to production facilities using the most up to date technologies, providing the highest quality for standard and special products and solutions.

(*) EC: International Electrotechnical Commission.

EN: European Standards, harmonizes all relevant European Standards NF、BS、VDE、DIN、CEI、UNE.

典型业绩

Reference



技术数据表

Technical data

SC (B) 10 系列变压器技术数据表(10 kV)
Technical data for SC (B) 10 series transformer (10 kV)

额定容量 Rated power (kVA)	电压组合 / Voltage combination			联结组别 Connection symbol	空载损耗 No-load loss (W)	负载损耗 On-load loss 75℃ (W)	空载电流 No-load current (%)	短路阻抗 Short circuit impedance (%)	声压级 Sound pressure (dB)	声功率级 Sound power (dB)	
	高压 HV (kV)	高压分接范围 Tapping range of HV (%)	低压 LV (kV)								
SC(B)10-100	6 6.3 6.6 10 10.5 11	±2*2.5%	0.4	Dyn11 或 or Yyn0	400	1370	1.5	4	44	57	
SC(B)10-125					470	1610	1.3				
SC(B)10-160					540	1860	1.1		45	58	
SC(B)10-200					620	2200			46	59	
SC(B)10-250					720	2410			47	60	
SCB10-315					880	3030	1.0		48	61	
SCB10-400					980	3480			49	62	
SCB10-500					1160	4260			50	63	
SCB10-630					1340	5130	0.85	51	64		
SCB10-630					1300	5200		6	51	64	
SCB10-800					1520	6070			52	65	
SCB10-1000					1770	7090			53	66	
SCB10-1250					2090	8400			54	67	
SCB10-1600					2450	10190			55	68	
SCB10-2000					3050	12500			0.7	55	69
SCB10-2500					3600	14900				56	70

*额定容量<315 kVA时，型号可能为SC10。
Rated Power<315 kVA, the type can be SC10.

SC (B) 11 系列变压器技术数据表(10 kV)
Technical data for SC (B) 11 series transformer (10 kV)

额定容量 Rated power (kVA)	电压组合 / Voltage combination			联结组别 Connection symbol	空载损耗 No-load loss (W)	负载损耗 On-load loss 75 °C(W)	空载电流 No-load current (%)	短路阻抗 Short circuit impedance (%)	声压级 Sound pressure (dB)	声功率级 Sound power (dB)	
	高压 HV (kV)	高压分接范围 Tapping range of HV (%)	低压 LV (kV)								
SC(B)11-100	6 6.3 6.6 10 10.5 11	± 2*2.5%	0.4	Dyn11 或 or Yyn0	360	1370	1.5	4	44	57	
SC(B)11-125					420	1610	1.3				
SC(B)11-160					480	1860	1.1		45	58	
SC(B)11-200					550	2200			46	59	
SC(B)11-250					640	2410			47	60	
SCB11 -315					790	3030	1.0		48	61	
SCB11-400					880	3480			49	62	
SCB11-500					1040	4260			50	63	
SCB11-630					1200	5130	0.85	6	51	64	
SCB11-630					1170	5200			51	64	
SCB11-800					1360	6070			52	65	
SCB11-1000					1590	7090			53	66	
SCB11-1250					1880	8400			54	67	
SCB11-1600					2200	10190			55	68	
SCB11-2000					2740	12500			0.7	55	69
SCB11-2500					3240	14900				56	70

*额定容量<315 kVA时，型号可能为SC11。
Rated Power<315 kVA, the type can be SC11.

SC (B) 12 系列变压器技术数据表(10 kV)**Technical data for SC (B) 12 series transformer (10 kV)**

额定容量 Rated power (kVA)	电压组合 / Voltage combination			联结组别 Connection symbol	空载损耗 No-load loss (W)	负载损耗 On-load loss 75℃ (W)	空载电流 No-load current (%)	短路阻抗 Short circuit impedance (%)	声压级 Sound pressure (dB)	声功率级 Sound power (dB)				
	高压 HV (kV)	高压分接范围 Tapping range of HV (%)	低压 LV (kV)											
SC(B)12-100	6 6.3 6.6 10 10.5 11	± 2*2.5%	0.4	Dyn11 或 or Yyn0	320	1370	1.5	4	44	57				
SC(B)12-125					375	1610	1.3							
SC(B)12-160					430	1860	1.1				45	58		
SC(B)12-200					495	2200							46	59
SC(B)12-250					575	2410								
SCB12-315					705	3030	1.0				48	61		
SCB12-400					785	3480	49	62						
SCB12-500					930	4260			50	63				
SCB12-630					1070	5130					0.85	51	64	
SCB12-630					1040	5200								
SCB12-800					1215	6070								
SCB12-1000					1415	7090								
SCB12-1250					1670	8400								
SCB12-1600					1960	10190								
SCB12-2000					2440	12500	0.7	55	69					
SCB12-2500					2880	14900	56			70				

*额定容量<315 kVA时，型号可能为SC12。

Rated Power<315 kVA, the type can be SC12.

SCB10 系列变压器技术数据表(20 kV)**Technical data for SCB10 series transformer (20 kV)**

额定容量 Rated power (kVA)	电压组合 / Voltage combination			联结组别 Connection symbol	空载损耗 No-load loss (W)	负载损耗 On-load loss 75 °C(W)	空载电流 No-load current (%)	短路阻抗 Short circuit impedance (%)	声压级 Sound pressure (dB)	声功率级 Sound power (dB)
	高压 HV (kV)	高压分接范围 Tapping range of HV (%)	低压 LV (kV)							
SCB10-315	20 22	± 2*2.5%	0.4	Dyn11 或 or Yyn0	970	3550	1.3	6	50	63
SCB10-400					1150	4220	1.1		51	64
SCB10-500					1350	5040	1.0		52	65
SCB10-630					1530	5970			53	66
SCB10-800					1750	7200			54	67
SCB10-1000					2070	8480	0.85		54	68
SCB10-1250					2380	10000			56	69
SCB10-1600					2790	12000			56	70
SCB10-2000					3240	14200	0.7		57	71
SCB10-2500					3870	16800			58	72

技术数据表

Technical data

SCB10 系列变压器技术数据表(35 kV)

Technical data for SCB10 series transformer (35 kV)

额定容量 Rated power (kVA)	电压组合 / Voltage combination			联结组别 Connection symbol	空载损耗 No-load loss (W)	负载损耗 On-load loss 75℃ (W)	空载电流 No-load current (%)	短路阻抗 Short circuit impedance (%)	声压级 Sound pressure (dB)	声功率级 Sound power (dB)
	高压 HV (kV)	高压分接范围 Tapping range of HV (%)	低压 LV (kV)							
SCB10-315	35 38.5	± 2*2.5%	0.4	Dyn11 或 or Yyn0	1170	3930	1.3	6	51	64
SCB10-400					1370	4720	1.1		52	65
SCB10-500					1620	5790	1.0		53	66
SCB10-630					1860	6700			54	67
SCB10-800					2160	7950			55	68
SCB10-1000					2430	9060	0.75		56	69
SCB10-1250					2830	11000			57	70
SCB10-1600					3240	13400			58	71
SCB10-2000					3820	15800			59	73
SCB10-2500					4450	19000			60	74

SCBZ10 系列有载调压变压器技术数据表(10 kV)

SCBZ10 series onload tap changer transformer data sheet (10 kV)

额定容量 Rated power (kVA)	电压组合 / Voltage Combination			联结组别 Connection symbol	空载损耗 No-load loss (W)	负载损耗 On-load loss 75℃ (W)	空载电流 No-load current (%)	短路阻抗 Short circuit impedance (%)	声压级 Sound pressure (dB)	声功率级 Sound power (dB)
	高压 HV (kV)	高压分接范围 Tapping range of HV (%)	低压 LV (kV)							
SCBZ10-315	6 6.3 6.6 10 10.5 11	± 4*2.5%	0.4	Dyn11 或 or Yyn0	990	3140	1.1	4	48	61
SCBZ10-400					1120	3720	1.0		49	62
SCBZ10-500					1290	4550			50	63
SCBZ10-630					1490	5380	0.85	6	51	64
SCBZ10-630					1440	5540			51	64
SCBZ10-800					1710	6540			52	65
SCBZ10-1000					1980	7650	0.7		53	66
SCBZ10-1250					2340	9050			54	67
SCBZ10-1600					2720	10800			54	68
SCBZ10-2000					3420	13200	0.7		55	69
SCBZ10-2500					3960	15800			56	70

SC(B) 系列干式配电变压器能效 3级技术数据(空、负载损耗不允许有正偏差)**SC(B) series dry distribution transformer energy efficiency grade 3 (No positive deviation is allowed for loss)**

额定容量 Rated power (kVA) 型号 / Type	电压组合 / Voltage combination			联结组别 Connection symbol	空载损耗 No-load loss (W)	负载损耗 On-load loss 120°C(W)	空载电流 No-load current (%)	短路阻抗 Short circuit impedance (%)	声压级 Sound pressure (dB)
	高压 HV (kV)	高压分接范围 Tapping range of HV (%)	低压 LV (kV)						
SC(B)-100/10-NX3	10	± 2*2.5%	0.4	Dyn11	400	1570	1.5	4	44
SC(B)-125/10-NX3					470	1850	1.3		44
SC(B)-160/10-NX3					540	2130	1.3		45
SC(B)-200/10-NX3					620	2530	1.1		46
SC(B)-250/10-NX3					720	2760	1.1		47
SCB-315/10-NX3					880	3470	1.0		48
SCB-400/10-NX3					980	3990	1.0		49
SCB-500/10-NX3					1160	4880	1.0		50
SCB-630/10-NX3					1340	5880	0.85		51
SCB-630/10-NX3					1300	5960	0.85	6	51
SCB-800/10-NX3					1520	6960	0.85		52
SCB-1000/10-NX3					1770	8130	0.85		53
SCB-1250/10-NX3					2090	9690	0.85		54
SCB-1600/10-NX3					2450	11730	0.85		55
SCB-2000/10-NX3					3050	14450	0.7		55
SCB-2500/10-NX3					3600	17170	0.7		56

*额定容量<315 kVA时, 型号可能为SC。

Rated Power<315 kVA, the type can be SC.

SCB 系列干式配电变压器能效 2级技术数据(空、负载损耗不允许有正偏差)**SCB series dry distribution transformer energy efficiency grade 2 (No positive deviation is allowed for loss)**

额定容量 Rated power (kVA) 型号 / Type	电压组合 / Voltage combination			联结组别 Connection symbol	空载损耗 No-load loss (W)	负载损耗 On-load loss 120°C(W)	空载电流 No-load current (%)	短路阻抗 Short circuit impedance (%)	声压级 Sound pressure (dB)
	高压 HV (kV)	高压分接范围 Tapping range of HV (%)	低压 LV (kV)						
SCB-630/10-NX2	10	± 2*2.5%	0.4	Dyn11	1040	5960	0.85	6	51
SCB-800/10-NX2					1215	6960	0.85		52
SCB-1000/10-NX2					1415	8130	0.85		53
SCB-1250/10-NX2					1670	9690	0.85		54
SCB-1600/10-NX2					1960	11730	0.85		55
SCB-2000/10-NX2					2440	14450	0.7		55
SCB-2500/10-NX2					2880	17170	0.7		56

SCB 系列干式配电变压器能效 1级技术数据(空、负载损耗不允许有正偏差)**SCB series dry distribution transformer energy efficiency grade 1 (No positive deviation is allowed for loss)**

额定容量 Rated power (kVA) 型号 / Type	电压组合 / Voltage combination			联结组别 Connection symbol	空载损耗 No-load loss (W)	负载损耗 On-load loss 120°C(W)	空载电流 No-load current (%)	短路阻抗 Short circuit impedance (%)	声压级 Sound pressure (dB)
	高压 HV (kV)	高压分接范围 Tapping range of HV (%)	低压 LV (kV)						
SCB-630/10-NX1	10	± 2*2.5%	0.4	Dyn11	935	5365	0.85	6	51
SCB-800/10-NX1					1095	6265	0.85		52
SCB-1000/10-NX1					1275	7315	0.85		53
SCB-1250/10-NX1					1505	8720	0.85		54
SCB-1600/10-NX1					1765	10555	0.85		55
SCB-2000/10-NX1					2195	13005	0.7		55
SCB-2500/10-NX1					2590	15455	0.7		56

外形尺寸

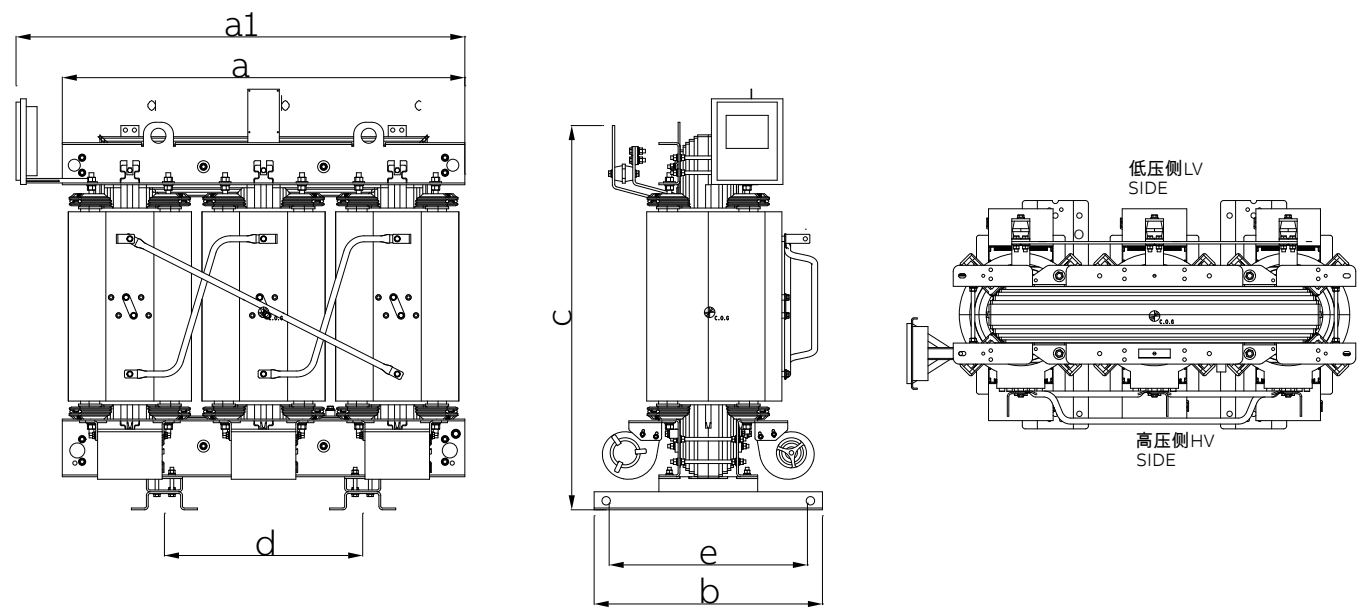
Outline dimension

SCB10 (11, 12) -630~2500/10/0.4 系列变压器外形尺寸及重量数据表(IP00)
Dimension and weight for SCB10 (11, 12) -630~2500/10/0.4 seriea transformer (IP00)

型号 / Modle	a	a1	b	c	d	e	重量 / Weight(kg)	图 / Fig.	
SCB10-630	1430	1585	760	1320	660	660	2070	1	5
SCB10-800	1550	1705	920	1335	820	820	2460		6
SCB10-1000	1550	1705		1425			2770		
SCB10-1250	1670	1825		1450			3330		7
SCB10-1600	1700	1855		1620			3950		
SCB10-2000	1880	2035	1170	1710	1070	1070	5000	8	8
SCB10-2500	1940	2095		1870			6050		

*以上尺寸适用于符合GB1094.11 (IEC60076.11) 标准正常使用环境条件的变压器
* The above dimensions apply to the transformers that comply with the GB1094.11 (IEC60076.11) standard and operate in normal conditions

VCC 干式变压器IP00外形图
Outline of VCC with IP00 enclosure



图Fig.1 SCB10-630~2500 kVA

低压接线端子图
LV terminal drawing

5

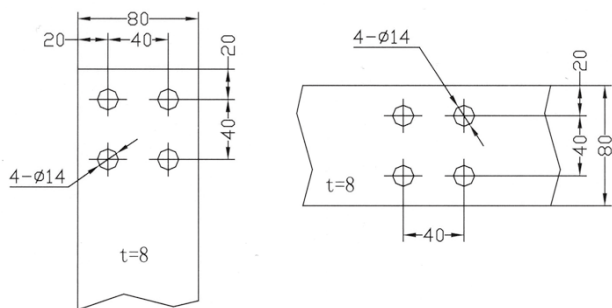


图 Fig.5(o, a, b, c)

6

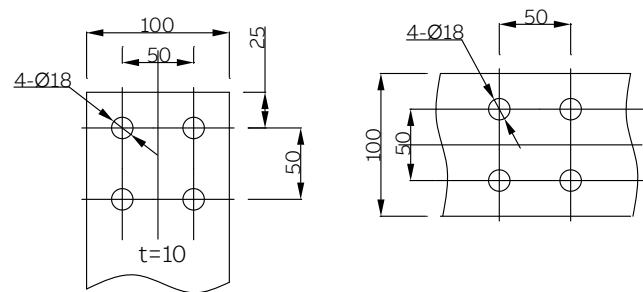


图 Fig.6(o, a, b, c)

7

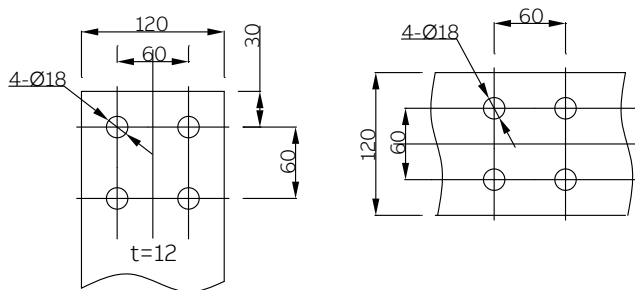


图 Fig.7(o, a, b, c)

8

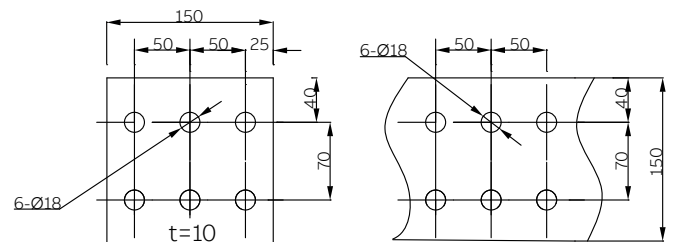


图 Fig.8(o, a, b, c)

外形尺寸

Outline dimension

SCB10（11，12）-630~2500/10/0.4 系列变压器外形尺寸及重量数据表(IP20)
Dimension and weight for SCB10 (11, 12) -630~2500/10/0.4 seriea transformer (IP20)

型号 / Modle	a	b	c	d	e	重量 / Weight (kg)	图 / Fig.	
SCB10-630	1800	1400	1700	660	1240	2320	2	5
SCB10-800	1900		1700	820		2720		
SCB10-1000	1900		1800			3040		
SCB10-1250	2000	1500	1800	1070	1340	3610	3	6
SCB10-1600	2000		2000			4250		
SCB10-2000	2200		2100			5370		
SCB10-2500	2300	1600	2300		1440	6530	4	7
								8

*以上尺寸适用于符合GB1094.11 (IEC60076.11) 标准正常使用环境条件的变压器
* The above dimensions apply to the transformers that comply with the GB1094.11 (IEC60076.11) standard and operate in normal conditions

VCC干式变压器IP20外形图
Outline of VCC with IP20 enclosure

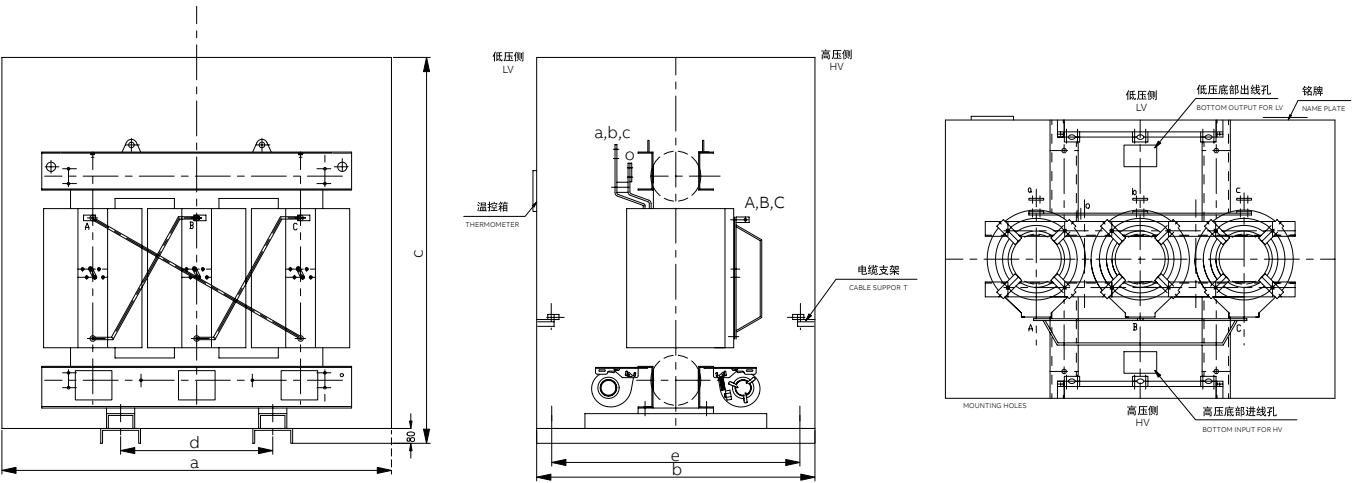


图2 SCB10 (11, 12) - 630~2500 kVA 下进下出
Fig.2 SCB10 (11, 12) - 630~2500 kVA bottm in-bottm out

VCC干式变压器IP20外形图

Outline of VCC with IP20 enclosure

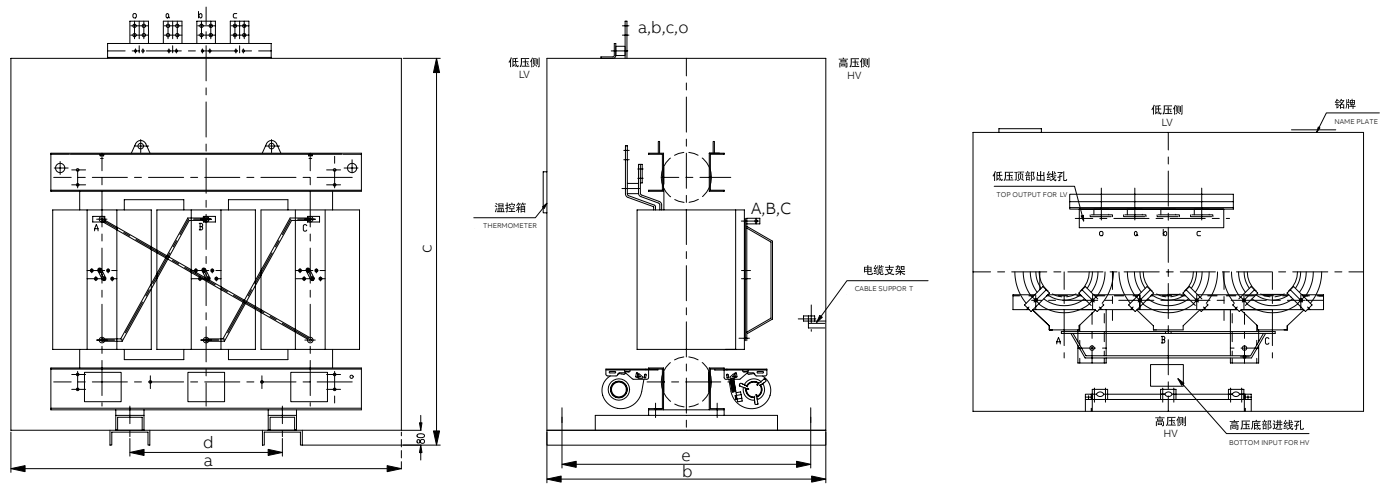


图3 SCB10 (11, 12) - 630~2500 kVA 下进上出

Fig.3 SCB10 (11, 12) - 630~2500 kVA bottm in-top out

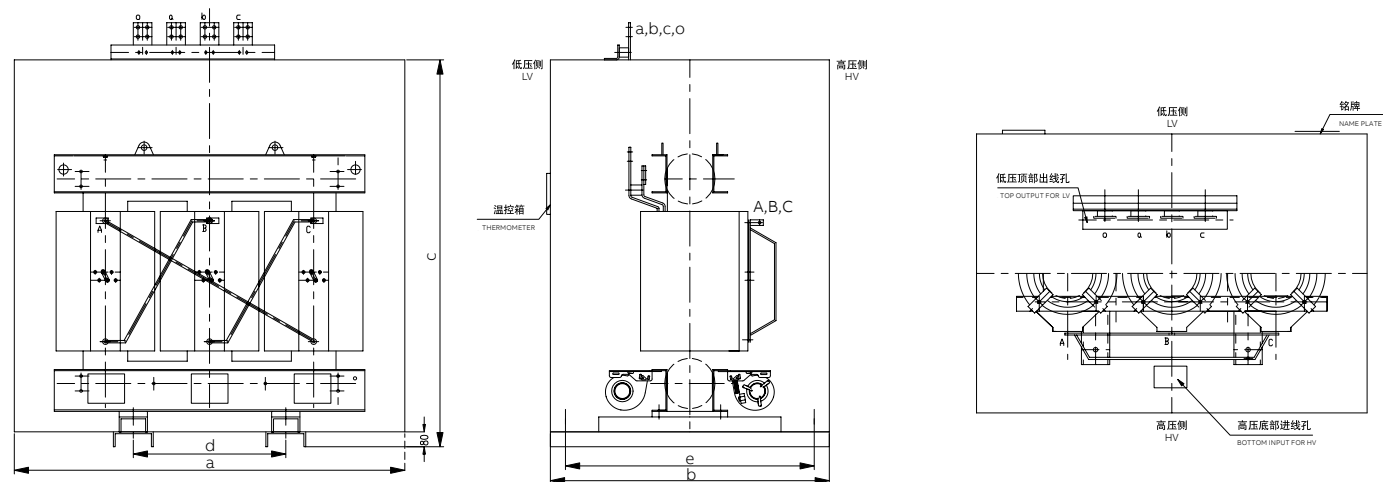


图4 SCB10 (11, 12) - 630~2500 kVA 上进上出

Fig.4 SCB10 (11, 12) - 630~2500 kVA top in-top out

试验报告

Test report

No. DY100959

MA 2009002873Z
AL (2009)年认证证书(348)号
CNAS L0153
MA 2009002873Z
AL (2009)年认证证书(348)号

检 验 报 告

Test Report

产品名称 干式变压器

型号规格 SCB10-400/10

受检单位 上海 ABB 变压器有限公司

试验类别 委托

产品质量监督检验研究院
Guangdong Testing Institute of Product Quality Supervision
国家中低压输配电设备质量监督检验中心
China National Quality Supervision and Testing Center for Medium and Low Voltage Transmission and Distribution Products
2010年11月11日

MA 2012002878Z
CNAS L11020
AL (2012)年认证证书(347)号
CQC

实验室名称: 国家电器产品质量监督检验中心
Lab Name: China National Center for Quality Supervision and Test of Electrical Apparatus Products

No. 13W0327-S

检 验 (试 验) 报 告

Test Report

委托单位: 上海 ABB 变压器有限公司
Client:

产品名称: 干式变压器
Name of Product:

产品型号: SCB10-800/10
Product Type:

检验类别: 委托试验
Test Category:

本实验室对出具的检验(试验)结果负责, 未经实验室书面同意, 不得部分地复制本报告。
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ILAC-MRA
CNAS L0153 (2009)年认证证书(348)号
AL 2009002873Z
MA

试验合格证书

No: CNCE/T-11101
(试验报告编号: DY110155)

生产单位: 上海 ABB 变压器有限公司

样品名称: 干式变压器

样品型号: SCB10-2500/10

样品序号: 1LCTV101103501

试验项目: 例行试验、温升试验、声级测定、短路承受能力试验、雷电冲击试验

试验依据: GB 1094.1-1996、GB 1094.2-1996、GB 1094.3-2003、GB 1094.5-2008、GB/T 1094.10-2003、GB 1094.11-2007、GB/T 10228-2008、委托技术要求

试验结论: 干式变压器(型号: SCB10-2500/10)例行试验、温升试验、声级测定、短路承受能力试验、雷电冲击试验的试验结果符合试验依据和委托技术要求, 样品上述试验合格。

发证日期: 2011年03月03日

批准: 林志力

国家中低压输配电设备质量监督检验中心
证书查询网: www.cnce.asia

ILAC-MRA
CNAS L0153 (2009)年认证证书(348)号
AL 2009002873Z
MA

试验合格证书

No: CNCE/T-11376
(试验报告编号: DY111005)

生产单位: 上海 ABB 变压器有限公司

样品名称: 干式变压器

样品型号: SCB11-2500/20

样品序号: 1LCTV110804001

试验项目: 例行试验、温升试验、雷电冲击试验

试验依据: GB 1094.1-1996、GB 1094.2-1996、GB 1094.3-2003、GB 1094.11-2007、GB/T 10228-2008、委托技术要求

试验结论: 干式变压器(型号: SCB11-2500/20)例行试验、温升试验、雷电冲击试验的试验结果符合试验依据和委托技术要求, 样品上述试验合格。

发证日期: 2011年09月05日

批准: 林志力

国家中低压输配电设备质量监督检验中心
证书查询网: www.cnce.asia






2012002878Z

实验室名称: 国家电器产品质量监督检验中心
Lab Name: China National Center for Quality Supervision and Test of Electrical Apparatus Products

No. 12N0229-S

检验 (试验) 报告 Test Report

委托单位: 上海 ABB 变压器有限公司
Client:

产品名称: 干式变压器
Name of Product:

产品型号: SCB10-2000/35
Product Type:

检验类别: 委托试验
Test Category:

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2009002878Z

实验室名称: 国家电器产品质量监督检验中心
Lab Name: China National Center for Quality Supervision and Test of Electrical Apparatus Products

No. 12N0015-S

检验 (试验) 报告 Test Report

委托单位: 上海 ABB 变压器有限公司
Client:

产品名称: 干式变压器
Name of Product:

产品型号: SCB10-1000/6
Product Type:

检验类别: 委托试验
Test Category:

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2009000149Z

检 验 报 告

试品型号及名称: ZQSCB10-3300/35
干式牵引整流变压器

委托单位: 上海 ABB 变压器有限公司

检验类别: 型式试验、例行试验、特殊试验

西安高压电器研究院有限公司实验认证中心
国家高压电器质量监督检验中心





2009002873Z

检 验 报 告 Test Report

产品名称: 三相干式变压器

型号规格: DTE5000/35

委托单位: 上海 ABB 变压器有限公司

检验类别: 委托

国家中低压输配电设备质量监督检验中心
China National Quality Supervision and Testing Center for Medium and Low Voltage Transmission and Distribution Products

2011年10月15日

试验报告

Test report

中国节能产品认证证书

证书编号: CQC14701109009

申请人名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

品牌: ABB

制造商名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

生产企业名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

产品名称: 干式变压器

系列/规格/型号: SCB-432/10-951 SCB-800/10-951 SCB-1000/10-951 SCB-1250/10-951

认证模式: 产品型式试验+初次工厂检查+获证后监督

产品标准/技术要求: GB20052-2013

上述产品符合CQC31-461212-2013认证规则的要求。

证书有效期: 2014年05月27日 至 2017年05月27日

证书有效期内本证书的有效性依据发证机构的定期监督获得保持。



主任: 张博



中国质量认证中心

中国 北京 南四环西路188号9区 100070
http://www.cqc.com.cn

JC 0073820

CHINA CERTIFICATE For ENERGY CONSERVATION PRODUCT

Certificate No: CQC14701109009

NAME AND ADDRESS OF THE APPLICANT: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

BRAND NAME: ABB

NAME AND ADDRESS OF THE MANUFACTURER: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

NAME AND ADDRESS OF THE FACTORY: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

PRODUCT NAME: Dry-type transformer

PRODUCT MODEL AND SPECIFICATION: SCB-432/10-951 SCB-800/10-951 SCB-1000/10-951 SCB-1250/10-951

CERTIFICATION MODEL: Type Testing of Product + Initial Factory Inspection + Follow up Factory Inspection

THE STANDARD/TECHNICAL REQUIREMENTS FOR THE PRODUCTS: GB20052-2013

This is to certify that the above-mentioned product meets the requirements of CQC31-461212-2013.

Valid dates of this Certificate: From May, 27, 2014 to May, 27, 2017

The validity of the certificate depend on the follow up inspection by the certification body at regular intervals.



President: 张博



CHINA QUALITY CERTIFICATION CENTRE

Section 9, No.188, Nanshuan Xilu, Beijing 100070 P.R.China
http://www.cqc.com.cn

JC 0073820

中国节能产品认证证书

证书编号: CQC14701109011

申请人名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

品牌: ABB

制造商名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

生产企业名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

产品名称: 干式变压器

系列/规格/型号: SCB-1800/10-951 SCB-2000/10-951 SCB-2500/10-951

认证模式: 产品型式试验+初次工厂检查+获证后监督

产品标准/技术要求: GB20052-2013

上述产品符合CQC31-461212-2013认证规则的要求。

证书有效期: 2014年05月27日 至 2017年05月27日

证书有效期内本证书的有效性依据发证机构的定期监督获得保持。



主任: 张博



中国质量认证中心

中国 北京 南四环西路188号9区 100070
http://www.cqc.com.cn

JC 0073821

CHINA CERTIFICATE For ENERGY CONSERVATION PRODUCT

Certificate No: CQC14701109011

NAME AND ADDRESS OF THE APPLICANT: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

BRAND NAME: ABB

NAME AND ADDRESS OF THE MANUFACTURER: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

NAME AND ADDRESS OF THE FACTORY: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

PRODUCT NAME: Dry-type transformer

PRODUCT MODEL AND SPECIFICATION: SCB-1800/10-951 SCB-2000/10-951 SCB-2500/10-951

CERTIFICATION MODEL: Type Testing of Product + Initial Factory Inspection + Follow up Factory Inspection

THE STANDARD/TECHNICAL REQUIREMENTS FOR THE PRODUCTS: GB20052-2013

This is to certify that the above-mentioned product meets the requirements of CQC31-461212-2013.

Valid dates of this Certificate: From May, 27, 2014 to May, 27, 2017

The validity of the certificate depend on the follow up inspection by the certification body at regular intervals.



President: 张博



CHINA QUALITY CERTIFICATION CENTRE

Section 9, No.188, Nanshuan Xilu, Beijing 100070 P.R.China
http://www.cqc.com.cn

JC 0073821



中国节能产品认证证书

证书编号: CQC14701109013

申请人名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

品牌: ABB

制造商名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

生产企业名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

产品名称: 干式变压器

系列/规格/型号: SCB-432/10-302 SCB-400/10-302 SCB-400/10-302 SCB-1250/10-302

认证模式: 产品型式试验+初次工厂检查+获证后监督

产品标准/技术要求: GB20052-2013

上述产品符合CQC31-461212-2013认证规则的要求。

证书有效期: 2014年05月27日 至 2017年05月27日

证书有效期内本证书的有效性依据发证机构的定期监督获得保持。



主任: 王长博



中国质量认证中心

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JC 0073822



CHINA CERTIFICATE For ENERGY CONSERVATION PRODUCT

Certificate No: CQC14701109013

NAME AND ADDRESS OF THE APPLICANT: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

BRAND NAME: ABB

NAME AND ADDRESS OF THE MANUFACTURER: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

NAME AND ADDRESS OF THE FACTORY: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

PRODUCT NAME: Dry-type transformer

PRODUCT MODEL AND SPECIFICATION: SCB-432/10-302 SCB-400/10-302 SCB-400/10-302 SCB-1250/10-302

CERTIFICATION MODEL: Type Testing of Product + Initial Factory Inspection + Follow up Factory Inspection

THE STANDARD/TECHNICAL REQUIREMENTS FOR THE PRODUCTS: GB20052-2013

This is to certify that the above-mentioned product meets the requirements of CQC31-461212-2013.

Valid dates of this Certificate: From May. 27, 2014 to May. 27, 2017

The validity of the certificate depend on the follow up inspection by the certification body at regular intervals.



President: 王长博



CHINA QUALITY CERTIFICATION CENTRE

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http://www.cqc.com.cn

JC 0073822



中国节能产品认证证书

证书编号: CQC14701109014

申请人名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

品牌: ABB

制造商名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

生产企业名称及地址: 上海ABB变压器有限公司
上海市浦东新区中江路2300号

产品名称: 干式变压器

系列/规格/型号: SCB-1600/10-302 SCB-2000/10-302 SCB-2500/10-302

认证模式: 产品型式试验+初次工厂检查+获证后监督

产品标准/技术要求: GB20052-2013

上述产品符合CQC31-461212-2013认证规则的要求。

证书有效期: 2014年05月27日 至 2017年05月27日

证书有效期内本证书的有效性依据发证机构的定期监督获得保持。



主任: 王长博



中国质量认证中心

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http://www.cqc.com.cn

JC 0073823



CHINA CERTIFICATE For ENERGY CONSERVATION PRODUCT

Certificate No: CQC14701109014

NAME AND ADDRESS OF THE APPLICANT: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

BRAND NAME: ABB

NAME AND ADDRESS OF THE MANUFACTURER: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

NAME AND ADDRESS OF THE FACTORY: ABB SHANGHAI TRANSFORMER CO.,LTD.
No. 2300 SHENJIANG ROAD,PUDONG DISTRICT,SHANGHAI

PRODUCT NAME: Dry-type transformer

PRODUCT MODEL AND SPECIFICATION: SCB-1600/10-302 SCB-2000/10-302 SCB-2500/10-302

CERTIFICATION MODEL: Type Testing of Product + Initial Factory Inspection + Follow up Factory Inspection

THE STANDARD/TECHNICAL REQUIREMENTS FOR THE PRODUCTS: GB20052-2013

This is to certify that the above-mentioned product meets the requirements of CQC31-461212-2013.

Valid dates of this Certificate: From May. 27, 2014 to May. 27, 2017

The validity of the certificate depend on the follow up inspection by the certification body at regular intervals.



President: 王长博



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联系我们

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